

# ORION - Technical Report

*Operational Resource Intelligence & Optimization Network. Generated from stored experiment artifacts.*

## 1. Executive Summary

ORION allocates limited operational resources - teams, vehicles and equipment - to time-sensitive work under capability, availability, capacity, travel, deadline and dependency constraints. It answers two questions a static schedule cannot: what is the best plan available now, and what is the cheapest way to recover when reality changes.

The engine is five solvers behind one shared objective, so their results are directly comparable. Two are exact or near-exact and report proven optimality where they have it; two are relaxations that say so; one is a greedy constructive heuristic and one an ALNS local search. Every plan is re-scored centrally by an independent scorer, because a solver's own objective is not a fair basis for comparison.

On the disruption demo the plan is built, a vehicle fails mid-shift, 7 affected tasks are identified, and two recovery strategies are run from the same disrupted state. Local repair recovers 92.5% of the lost service in 0.090 s changing 9 of 46 assignments; full re-optimization reaches 628.37 in 5.024 s - about 56x slower. The speed difference is the useful result; the quality difference is small, which is the point of measuring rather than assuming it.

Limitations are stated throughout and are not cosmetic. MIN\_COST\_FLOW is a transportation relaxation: it does not model route ordering or time windows, so it serves fewer customers than the routing solvers and is never presented as their equal. The job-shop instances carry no published optima in the downloaded artifact, so none are claimed. Solver timeouts are reported as timeouts.

## 2. Engineering Question

This is the question the experiments are designed to answer. The secondary questions are narrower versions of it:

How does the gap between exact and approximate methods grow with problem size and constraint pressure?

How much solution quality is lost when the time budget tightens?

How quickly can a system recover from a disruption, and what does that cost in churn?

Which constraints cause the largest service-quality loss?

## 3. Why I Chose This Question

Constrained resource allocation is one of the harder problems in operational research, and it is hard for a specific reason: the decisions interact. Assigning a team to one job changes when it is available for the next, which changes whether that next job can meet its deadline, which changes the choice before it. Search space grows factorially, and the practical answer is rarely the global optimum.

A static schedule is also not what operations actually run. Vehicles break down, teams are re-tasked, weather delays a delivery and an urgent job arrives at 07:00 that was not in the morning plan. A system that produces a good plan once and stops is not a planning system; the interesting question is how it behaves when the plan stops matching the world.

That makes recovery speed a first-class requirement, not a nice-to-have. Minutes of recovery are operationally real, and a plan that is 1% better but takes an hour to produce may be worth less than a plan that is 3% worse and arrives in ten seconds. ORION measures both, and reports the trade-off rather than declaring a winner.

Exact and approximate methods have genuinely different trade-offs, and the honest way to present that is with measured evidence. ORION therefore runs a brute-force reference on small instances where the optimum is computable - 7 fixtures, 0 mismatches against the proven optimum, 0 solvers scoring above it - so the approximate methods' gaps are measured against truth rather than against each other.

Problem size and constraint pressure both grow faster than any solver's ability to prove optimality.

Exact methods do not scale here on their own: CP-SAT hit its time limit in 20% of scalability runs.

Relaxations are fast and incomplete, and their incompleteness is measurable - a relaxation reporting a higher objective is not a better solver, it is a different problem.

Operational conditions change, so recovery behaviour is a quality dimension of the system, not an operational detail.

## 4. System Overview

The system is a library plus a thin product shell. The optimization core has no dependency on the web layer, the database or the CLI, which is what allows the same code to be exercised by a unit test, a benchmark run and an HTTP request without behaving differently in each.

orion.domain entities, plans, events, travel  
orion.optimization objective, models, five solvers, reference checker  
orion.planning planner, replanner, comparison, what-if  
orion.simulation discrete-event engine, disruption generator  
orion.evaluation metrics, failure taxonomy  
orion.experiments immutable experiment store, benchmark suites  
orion.data loaders, adapters, exporters  
orion.evidence the single reader every document uses  
orion.report report generation  
orion.cli command line entry point

Evidence in this report comes from 5 live experiment suites at commit(s) 5fe96854, dc70e26f, with 33 superseded runs preserved but not quoted.

## 5. Domain Model

A scenario is the immutable input: a set of Task objects with a release time, a deadline, a duration, a priority, required capabilities and optional predecessor tasks; a set of Resource objects with capabilities, a work calendar, a capacity mapping, a home location and a speed factor; a TravelMatrix; and a ScenarioConstraints record that switches each constraint on or off explicitly.

An Assignment is the unit of a plan: one task on one resource with a start, an end, a travel distance and a lateness. A Plan is a set of assignments plus the objective breakdown, the solver runs that produced it and the validation findings. Keeping solver runs on the plan rather than flattening them into plan fields is what makes it possible to report a solver's real status when the plan itself was assembled by a different layer.

| Scenario property | Value                                                                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 60                | 60 tasks, 14 resources (9 teams / 5 vehicles)                                                                                                                              |
| horizon           | horizon [00:00-12:00) demand 3169 min available 4241 min                                                                                                                   |
| capabilities      | capabilities ['crane_heavy', 'electrical', 'hazmat', 'heavy_lift', 'hvac', 'network', 'plumbing', 'refrigerated', 'structural', 'transport', 'vehicle', 'water_treatment'] |
| static            | static feasibility: FEASIBLE                                                                                                                                               |

## 6. Mathematical Formulation

One objective, defined once in orion.optimization.objective and applied to every solver. It is a weighted sum of service terms and penalty terms:

maximise sum over assigned tasks of  $w_{\text{completion}} * \text{value}(\text{task})$   
+  $w_{\text{priority}} * \text{priority\_bonus}(\text{task})$   
-  $w_{\text{travel}} * \text{km}$   
-  $w_{\text{cost}} * \text{operating cost}$   
-  $w_{\text{late}} * \text{late minutes} / 60$   
-  $w_{\text{dispatch}} * \text{dispatches per resource}$   
-  $w_{\text{overload}} * \text{overload minutes}$   
-  $w_{\text{move\_idle}} * \text{unnecessary moves}$   
-  $w_{\text{violation}} * \text{hard constraint violations}$

The objective is a reward: higher is better. That sign convention is enforced in the reference checker, where an early inversion in the gap calculation reported a solver as better than the proved optimum when it was worse.

Objective weights used in the recorded runs

### *Why the rescoring matters*

Each solver has its own internal objective - CP-SAT works in scaled integers, MILP prices travel in raw kilometres, the flow relaxation prices assignment arcs. Quoting those side by side would be meaningless. So `run_solver` re-scores every returned plan with one independent scorer and keeps the solver's own figure separately as `internal_objective`. Every objective in this report is that common score.

## 7. Constraint System

| Constraint                       | Where enforced                                       | On failure                          |
|----------------------------------|------------------------------------------------------|-------------------------------------|
| Capability / eligibility         | context pair enumeration; every solver               | task cannot be assigned             |
| Work calendar / availability     | <code>schedule_sequence</code>                       | sequence rejected; prefix kept      |
| Capacity (per capability)        | <code>schedule_sequence</code> , CP-SAT, MILP        | overload penalty and hard violation |
| Time window (release / deadline) | <code>schedule_sequence</code> , all solvers         | late penalty; miss is a violation   |
| Travel and return to depot       | <code>TravelMatrix</code> , every solver             | infeasible insertion                |
| Dependencies (cross-resource)    | <code>dependency_bounds</code> , shared materialiser | task delayed or dropped             |
| Maintenance blocks               | domain events, applied to availability               | resource unavailable for the window |

Dependencies are the constraint that caused the most reconstruction bugs, and the reason is structural: a job's operations are spread across different resources, so no single resource's task list contains its own chain. Correctness therefore depends on a global bound derived from every other resource's finish times - not on any per-resource ordering, which is why the same materialiser can serve a solver that knows its model's start ordering and one that does not.

Hard violations are never silently repaired. A plan that cannot be materialised into a schedule satisfying these constraints is reported as ERROR, and the benchmark records that rather than scoring the invalid plan.

## 8. Solver Architecture

| Solver        | Method                            | Optimality claim                       | Principal limitation                                     |
|---------------|-----------------------------------|----------------------------------------|----------------------------------------------------------|
| CP-SAT        | exact constraint programming      | proven optimal when it reports OPTIMAL | finds nothing inside short budgets at scale              |
| MILP          | linear program, big-M precedence  | relaxation optimal                     | precedence is relaxed, so its makespan can be optimistic |
| MIN_COST_FLOW | transportation relaxation         | relaxation optimal                     | no route order, no time windows                          |
| HEURISTIC     | greedy density insertion          | approximate                            | fast, leaves work unplaced under pressure                |
| LOCAL_SEARCH  | ALNS / large-neighbourhood search | approximate                            | slower; still incomplete                                 |

### ***Exactness is verified, not asserted***

A brute-force enumerator computes the true optimum on 7 hand-built micro-instances. Across all five solvers there were 0 mismatches against a proven optimum and 0 cases of a solver scoring above one - a negative result that matters, because a solver that beats the optimum is a bug in the scorer, not a good solver.

| Heuristic    | Fixtures | Mean gap vs optimum | Mean runtime (s) |
|--------------|----------|---------------------|------------------|
| HEURISTIC    | 7        | -0.1%               | 0.0002           |
| LOCAL_SEARCH | 7        | 0.0%                | 0.0453           |

7 instances showed a shortfall against the reference, the worst at -3.70%. Those belong to the relaxations and the heuristics, and they are left in the record.

## 9. Heuristics

The constructive heuristic sorts work by value per minute and inserts each task at the cheapest feasible position across all eligible resources. The local search wraps that in an adaptive large-neighbourhood search: destroy and repair operators, a record-to-record acceptance criterion and a temperature schedule, with the time budget as the only stopping rule.

Both are seeded, and seed sensitivity is measured rather than assumed. On the reference fixtures both produced a spread of exactly zero across five seeds, which is the honest observation: at that size the search space is too small for the stochastic components to matter.

| Heuristic    | Seeds | Mean objective | Spread | Mean runtime (s) |
|--------------|-------|----------------|--------|------------------|
| HEURISTIC    | 5     | 67.0200        | 0.0000 | 0.0005           |
| LOCAL_SEARCH | 5     | 67.4067        | 0.0000 | 0.0933           |

One heuristic defect is worth recording because it was invisible from the outside. The insertion pool is ordered by value density, not by dependency, so a high-value successor could be inserted ahead of its own prerequisite. The pool order looked reasonable and the solver reported success every time; only an independent validator that re-read the instance's own precedence table found the violation. The fix restricts which positions are legal rather than refusing insertions, because refusing them stranded whole jobs - the first attempt at the fix placed zero operations on every instance.

## 10. Simulation Engine

The simulation is a discrete-event engine over the plan, not an animation of it. Events are typed and ordered, each one applied exactly once, and the engine's own state is what the replanner reads after a disruption. That distinction matters: a visualisation that replays a recorded plan cannot tell you what the system believes at 07:14, and the recovery question is entirely about that.

| Observed      | Value                                                                             |
|---------------|-----------------------------------------------------------------------------------|
| sim 0-720 (00 | sim 0-720 (00:00-12:00): 46 completed, 11 late, 0 failed, 0 replan(s), 138 events |
| first events  | first events: 06:36 TASK_ARRIVE   06:42 TASK_ARRIVE   06:50 TASK_ARRIVE           |

### *Determinism*

The engine is driven by an explicit clock and a seeded RNG, with no wall-clock reads. Two runs of the same plan produce the same event sequence, which is what makes the disruption benchmark comparable at all.

## 11. Disruption Model

Disruptions are first-class domain objects, not plan edits. Applying one produces a new state and a set of identified consequences, so the question 'what did this break' has a computed answer rather than a diff.

| Disruption           | Effect on state                                  | Consequence                           |
|----------------------|--------------------------------------------------|---------------------------------------|
| VEHICLE_FAILURE      | resource unavailable for a window                | its assigned tasks must be re-placed  |
| TEAM_UNAVAILABLE     | team loses capability coverage                   | tasks needing that capability at risk |
| MAINTENANCE_EVENT    | availability block, clipped to the work calendar | capacity removed in-window            |
| OP_AVAILABILITY_DROP | reduced availability                             | later start or dropped work           |
| DEMAND_SURGE         | new urgent tasks appended                        | plan must absorb unscheduled work     |
| TRAVEL_INCREASE      | matrix scaled                                    | routes may become infeasible          |

Blocked windows are clipped to the resource's own work calendar. An unclipped block produced availability intervals longer than the shift existed, which then had to be handled defensively downstream - a symptom of the wrong representation, not of a missing guard.

## 12. Replanning

Two strategies run from the same disrupted state, so their difference is strategy, not inputs.

### **Local repair**

Tasks that the disruption actually affected are removed, the unaffected assignments are kept in their prior order, and the affected work is re-inserted where it now fits. Two defects made this materially worse than it should have been: survivors were sorted by task id rather than by their prior plan order, and a single failed sequence caused the whole resource's route to be discarded instead of its longest feasible prefix. Fixing both moved recovery from 57.1% to 83.8% on the demo scenario at the time of the change.

### **Full re-optimization**

The disrupted scenario is re-solved from scratch with no memory of the prior plan. It is the quality ceiling for the strategy, not necessarily the better operational choice.

| Strategy             | Objective | Runtime (s) | Churn     | Service recovery |
|----------------------|-----------|-------------|-----------|------------------|
| Local repair         | 629.20    | 0.090       | 9/46      | -                |
| Full re-optimization | 628.37    | 5.024       | full plan | 92.5%            |

Local repair is about 56x faster and within 0.84 objective of the full re-optimization, while changing a small fraction of the plan. Neither strategy is universally better: local repair preserves plan stability, which matters when a plan has already been communicated, and full re-optimization wins when the disruption invalidates most of the plan.

## **13. Public Benchmarks**

### **Sources**

| Dataset                       | Source                                             | License                     | Use                                                   |
|-------------------------------|----------------------------------------------------|-----------------------------|-------------------------------------------------------|
| Solomon VRPTW (100 customers) | SINTEF / Solomon, via the OR-Library-style archive | academic use, public domain | vehicle routing with time windows                     |
| OR-Library job shop           | Lawrence, M. and Watson, R. (1997), jobshop1.txt   | public academic archive     | job-shop scheduling, machine and precedence structure |

### **Mapping and assumptions**

Solomon DUE DATE bounds service start, not completion. Reading it as a completion bound reported a median lateness of 33 minutes that was an artifact of the mapping.

Solomon's capacity constraint is dropped: ORION's resources are capability- and time-bounded rather than load-bounded.

Job-shop machines become resources with machine-specific capabilities, so each operation is eligible for exactly one machine. An earlier version gave every machine the shared capability "machine", which made all machines eligible for all operations - a machine-free pool wearing a job-shop label.

The job-shop horizon is at least the heavier of the longest single job and the heaviest machine load, and no more than the total work. It had been twice the longest job, which for ft10 gave 1310 against a genuine lower bound of 655; CP-SAT's first solution at 1309 was pinned against a ceiling the adapter had invented.

### **Results**

| Instance | Solver | Method                         | Status             | Assigned | Objective | Late | Travel (km) | Runtime (s) |
|----------|--------|--------------------------------|--------------------|----------|-----------|------|-------------|-------------|
| ft10     | CP_SAT | exact (constraint programming) | OPTIMAL            | 100      | 1,400.00  | 0    | 0.00        | 0.168       |
| ft10     | MILP   | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 100      | 1,400.00  | 0    | 0.00        | 0.007       |

| Instance | Solver        | Method                       | Status         | Assigned | Objective | Late | Travel (km) | Runtime (s) |
|----------|---------------|------------------------------|----------------|----------|-----------|------|-------------|-------------|
| ft10     | HEURISTIC     | heuristic (greedy insertion) | FEASIBLE       | 63       | 882.00    | 0    | 0.00        | 0.107       |
| ft10     | LOCAL_SEARCH  | heuristic (ALNS improvement) | TIME_LIMIT     | 63       | 882.00    | 0    | 0.00        | 5.013       |
| ft10     | MIN_COST_FLOW | relaxation (transportation)  | NOT_APPLICABLE | 0        | 0.00      | 0    | 0.00        | 0.000       |

  

| Instance | Solver        | Method                         | Status             | Assigned | Objective | Late | Travel (km) | Runtime (s) |
|----------|---------------|--------------------------------|--------------------|----------|-----------|------|-------------|-------------|
| ft10     | CP_SAT        | exact (constraint programming) | OPTIMAL            | 100      | 1,400.00  | 0    | 0.00        | 0.170       |
| ft10     | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 100      | 1,400.00  | 0    | 0.00        | 0.007       |
| ft10     | HEURISTIC     | heuristic (greedy insertion)   | FEASIBLE           | 63       | 882.00    | 0    | 0.00        | 0.107       |
| ft10     | LOCAL_SEARCH  | heuristic (ALNS improvement)   | FEASIBLE           | 63       | 882.00    | 0    | 0.00        | 6.899       |
| ft10     | MIN_COST_FLOW | relaxation (transportation)    | NOT_APPLICABLE     | 0        | 0.00      | 0    | 0.00        | 0.000       |

  

| Instance | Solver        | Method                        | Status             | Assigned | Objective | Late | Travel (km) | Runtime (s) |
|----------|---------------|-------------------------------|--------------------|----------|-----------|------|-------------|-------------|
| c101     | LOCAL_SEARCH  | heuristic (ALNS improvement)  | TIME_LIMIT         | 99       | 1,371.90  | 0    | 705.00      | 5.028       |
| r101     | LOCAL_SEARCH  | heuristic (ALNS improvement)  | TIME_LIMIT         | 99       | 1,360.20  | 37   | 1,112.17    | 5.009       |
| c101     | HEURISTIC     | heuristic (greedy insertion)  | FEASIBLE           | 99       | 1,358.51  | 39   | 726.22      | 0.296       |
| r101     | HEURISTIC     | heuristic (greedy insertion)  | FEASIBLE           | 99       | 1,353.86  | 58   | 996.42      | 0.292       |
| c101     | MILP          | relaxation (big-M precedence) | RELAXATION_OPTIMAL | 81       | 999.81    | 62   | 1,454.42    | 1.422       |
| c101     | MIN_COST_FLOW | relaxation (transportation)   | RELAXATION_OPTIMAL | 46       | 536.76    | 32   | 668.25      | 0.007       |
| r101     | MILP          | relaxation (big-M precedence) | RELAXATION_OPTIMAL | 27       | 347.64    | 24   | 788.09      | 0.648       |
| r101     | MIN_COST_FLOW | relaxation (transportation)   | RELAXATION_OPTIMAL | 21       | 270.51    | 14   | 594.46      | 0.006       |

| Instance | Solver | Method                         | Status     | Assigned | Objective | Late | Travel (km) | Runtime (s) |
|----------|--------|--------------------------------|------------|----------|-----------|------|-------------|-------------|
| c101     | CP_SAT | exact (constraint programming) | TIME_LIMIT | 0        | 0.00      | 0    | 0.00        | 5.400       |
| r101     | CP_SAT | exact (constraint programming) | TIME_LIMIT | 0        | 0.00      | 0    | 0.00        | 5.143       |

  

| Instance | Solver        | Method                         | Status             | Assigned | Objective | Late | Travel (km) | Runtime (s) |
|----------|---------------|--------------------------------|--------------------|----------|-----------|------|-------------|-------------|
| c101     | LOCAL_SEARCH  | heuristic (ALNS improvement)   | TIME_LIMIT         | 99       | 1,372.25  | 0    | 687.48      | 30.107      |
| r101     | LOCAL_SEARCH  | heuristic (ALNS improvement)   | FEASIBLE           | 99       | 1,362.38  | 29   | 1,060.99    | 27.333      |
| c101     | HEURISTIC     | heuristic (greedy insertion)   | FEASIBLE           | 99       | 1,358.51  | 39   | 726.22      | 0.294       |
| r101     | CP_SAT        | exact (constraint programming) | FEASIBLE           | 99       | 1,357.32  | 14   | 1,409.02    | 30.258      |
| r101     | HEURISTIC     | heuristic (greedy insertion)   | FEASIBLE           | 99       | 1,353.86  | 58   | 996.42      | 0.288       |
| c101     | CP_SAT        | exact (constraint programming) | FEASIBLE           | 99       | 1,331.54  | 11   | 2,624.19    | 30.498      |
| c101     | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 81       | 999.81    | 62   | 1,454.42    | 0.878       |
| c101     | MIN_COST_FLOW | relaxation (transportation)    | RELAXATION_OPTIMAL | 46       | 536.76    | 32   | 668.25      | 0.007       |
| r101     | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 27       | 347.64    | 24   | 788.09      | 0.633       |
| r101     | MIN_COST_FLOW | relaxation (transportation)    | RELAXATION_OPTIMAL | 21       | 270.51    | 14   | 594.46      | 0.006       |

### ***No published optima are claimed***

The job-shop artifact as distributed does not contain verified optimal makespans, so this report states optimum: unknown for every job-shop instance rather than embedding best-known values that cannot be checked against the file. A table of makespans recalled from memory was removed from the adapter for exactly that reason: it looked like evidence and was not.

MIN\_COST\_FLOW's results deserve the explicit note the project requires: it is a transportation relaxation. On c101 at a 30-second budget it serves 46 of 99 customers. That is not a solver failure - it is the predicted behaviour of a model with no route ordering and no time windows, and the reason it is labelled RELAXATION\_OPTIMAL rather than OPTIMAL.

## **14. Experimental Method**

| Suite               | Rows | Commit   | Recorded (UTC)            | Seed |
|---------------------|------|----------|---------------------------|------|
| constraint_pressure | 20   | 5fe96854 | 2026-09-26T23:24:36+00:00 | 2026 |

| Suite             | Rows | Commit   | Recorded (UTC)            | Seed |
|-------------------|------|----------|---------------------------|------|
| disruption_stress | 81   | 5fe96854 | 2026-09-26T23:24:36+00:00 | 2026 |
| scalability       | 50   | dc70e26f | 2026-09-26T23:00:11+00:00 | 2026 |
| solver_comparison | 30   | dc70e26f | 2026-09-26T23:04:45+00:00 | 2026 |
| time_budget       | 25   | 5fe96854 | 2026-09-26T23:24:45+00:00 | 2026 |

## Immutability

Experiment directories are never overwritten. A run that turns out to be invalid is superseded: it keeps its directory, its rows and its manifest, gains a status and a recorded reason, and stops being readable as evidence. 33 runs are currently in that state.

## What makes a row quotable

The status comes from the solver run, not from the plan object, which assembles its own status field and can overwrite the solver's.

A row claiming a result must carry a non-zero objective and a real plan.

TIME\_LIMIT is a result, not a failure, and is never reclassified to make a table look tidier.

Relaxation-based solvers report RELAXATION\_OPTIMAL and never OPTIMAL.

Gap columns are computed against the best result at the same problem size and the same time budget.

## 15. Metrics

| Metric         | Definition                                    | Source             |
|----------------|-----------------------------------------------|--------------------|
| Objective      | the single shared service-minus-penalty score | independent scorer |
| Service level  | assigned tasks / released tasks               | plan               |
| Late tasks     | assignments finishing after their deadline    | plan               |
| Travel         | sum of assignment travel distance in km       | plan               |
| Operating cost | sum of per-resource operating cost            | plan               |
| Utilisation    | busy minutes / available minutes              | plan               |
| Runtime        | wall-clock seconds for the solver run         | solver run         |
| Churn          | assignments changed against the prior plan    | replanner          |
| Recovery       | service regained as a share of service lost   | replanner          |

Every solver's plan is scored by the same independent scorer. The solver's own internal figure is retained separately and never mixed into a comparison table, because the two are in different units - one of the defects fixed here was a flow relaxation double-charging travel that the scorer had already included, which made 88 of 99 customers look unprofitable.

## 16. Solver Results

| Datase t | Instanc e | Budg et | Solver    | Method                       | Status   | Objectiv e | Assigne d | Lat e | Travel (km) | Runtim e (s) |
|----------|-----------|---------|-----------|------------------------------|----------|------------|-----------|-------|-------------|--------------|
| solomon  | c101      | 5.0s    | HEURISTIC | heuristic (greedy insertion) | FEASIBLE | 1,358.51   | 99        | 39    | 726.22      | 0.296        |

| Dataset | Instance | Budget | Solver        | Method                         | Status             | Objective | Assigned | Lat | Travel (km) | Runtime (s) |
|---------|----------|--------|---------------|--------------------------------|--------------------|-----------|----------|-----|-------------|-------------|
| solomon | c101     | 30.0s  | HEURISTIC     | heuristic (greedy insertion)   | FEASIBLE           | 1,358.51  | 99       | 39  | 726.22      | 0.294       |
| solomon | c101     | 5.0s   | LOCAL_SEARCH  | heuristic (ALNS improvement)   | TIME_LIMIT         | 1,371.90  | 99       | 0   | 705.00      | 5.028       |
| solomon | c101     | 30.0s  | LOCAL_SEARCH  | heuristic (ALNS improvement)   | TIME_LIMIT         | 1,372.25  | 99       | 0   | 687.48      | 30.107      |
| solomon | c101     | 5.0s   | CP_SAT        | exact (constraint programming) | TIME_LIMIT         | 0.00      | 0        | 0   | 0.00        | 5.400       |
| solomon | c101     | 30.0s  | CP_SAT        | exact (constraint programming) | FEASIBLE           | 1,331.54  | 99       | 11  | 2,624.19    | 30.498      |
| solomon | c101     | 5.0s   | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 999.81    | 81       | 62  | 1,454.42    | 1.422       |
| solomon | c101     | 30.0s  | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 999.81    | 81       | 62  | 1,454.42    | 0.878       |
| solomon | c101     | 5.0s   | MIN_COST_FLOW | relaxation (transportation)    | RELAXATION_OPTIMAL | 536.76    | 46       | 32  | 668.25      | 0.007       |
| solomon | c101     | 30.0s  | MIN_COST_FLOW | relaxation (transportation)    | RELAXATION_OPTIMAL | 536.76    | 46       | 32  | 668.25      | 0.007       |
| solomon | r101     | 5.0s   | HEURISTIC     | heuristic (greedy insertion)   | FEASIBLE           | 1,353.86  | 99       | 58  | 996.42      | 0.292       |
| solomon | r101     | 30.0s  | HEURISTIC     | heuristic (greedy insertion)   | FEASIBLE           | 1,353.86  | 99       | 58  | 996.42      | 0.288       |
| solomon | r101     | 5.0s   | LOCAL_SEARCH  | heuristic (ALNS improvement)   | TIME_LIMIT         | 1,360.20  | 99       | 37  | 1,112.17    | 5.009       |
| solomon | r101     | 30.0s  | LOCAL_SEARCH  | heuristic (ALNS improvement)   | FEASIBLE           | 1,362.38  | 99       | 29  | 1,060.99    | 27.333      |
| solomon | r101     | 5.0s   | CP_SAT        | exact (constraint programming) | TIME_LIMIT         | 0.00      | 0        | 0   | 0.00        | 5.143       |
| solomon | r101     | 30.0s  | CP_SAT        | exact (constraint programming) | FEASIBLE           | 1,357.32  | 99       | 14  | 1,409.02    | 30.258      |
| solomon | r101     | 5.0s   | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 347.64    | 27       | 24  | 788.09      | 0.648       |
| solomon | r101     | 30.0s  | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 347.64    | 27       | 24  | 788.09      | 0.633       |

| Datase t | Instanc e | Budg et | Solver        | Method                         | Status             | Objectiv e | Assigne d | Lat e | Travel (km) | Runtim e (s) |
|----------|-----------|---------|---------------|--------------------------------|--------------------|------------|-----------|-------|-------------|--------------|
| solomon  | r101      | 5.0s    | MIN_COST_FLOW | relaxation (transportation)    | RELAXATION_OPTIMAL | 270.51     | 21        | 14    | 594.46      | 0.006        |
| solomon  | r101      | 30.0s   | MIN_COST_FLOW | relaxation (transportation)    | RELAXATION_OPTIMAL | 270.51     | 21        | 14    | 594.46      | 0.006        |
| jobshop  | ft10      | 5.0s    | HEURISTIC     | heuristic (greedy insertion)   | FEASIBLE           | 882.00     | 63        | 0     | 0.00        | 0.107        |
| jobshop  | ft10      | 30.0s   | HEURISTIC     | heuristic (greedy insertion)   | FEASIBLE           | 882.00     | 63        | 0     | 0.00        | 0.107        |
| jobshop  | ft10      | 5.0s    | LOCAL_SEARCH  | heuristic (ALNS improvement)   | TIME_LIMIT         | 882.00     | 63        | 0     | 0.00        | 5.013        |
| jobshop  | ft10      | 30.0s   | LOCAL_SEARCH  | heuristic (ALNS improvement)   | FEASIBLE           | 882.00     | 63        | 0     | 0.00        | 6.899        |
| jobshop  | ft10      | 5.0s    | CP_SAT        | exact (constraint programming) | OPTIMAL            | 1,400.00   | 100       | 0     | 0.00        | 0.168        |
| jobshop  | ft10      | 30.0s   | CP_SAT        | exact (constraint programming) | OPTIMAL            | 1,400.00   | 100       | 0     | 0.00        | 0.170        |
| jobshop  | ft10      | 5.0s    | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 1,400.00   | 100       | 0     | 0.00        | 0.007        |
| jobshop  | ft10      | 30.0s   | MILP          | relaxation (big-M precedence)  | RELAXATION_OPTIMAL | 1,400.00   | 100       | 0     | 0.00        | 0.007        |
| jobshop  | ft10      | 5.0s    | MIN_COST_FLOW | relaxation (transportation)    | NOT_APPLICABLE     | 0.00       | 0         | 0     | 0.00        | 0.000        |
| jobshop  | ft10      | 30.0s   | MIN_COST_FLOW | relaxation (transportation)    | NOT_APPLICABLE     | 0.00       | 0         | 0     | 0.00        | 0.000        |

## 17. Scalability

| Tasks | Budget (s) | Solver        | Status             | Assigned | Objective | Runtime (s) | Gap to best |
|-------|------------|---------------|--------------------|----------|-----------|-------------|-------------|
| 10    | 5          | CP_SAT        | OPTIMAL            | 9        | 128.31    | 0.021       | 9.11%       |
| 10    | 5          | HEURISTIC     | FEASIBLE           | 10       | 141.17    | 0.001       | 0.00%       |
| 10    | 5          | LOCAL_SEARCH  | FEASIBLE           | 10       | 141.17    | 0.116       | 0.00%       |
| 10    | 5          | MILP          | RELAXATION_OPTIMAL | 7        | 98.65     | 0.003       | 30.12%      |
| 10    | 5          | MIN_COST_FLOW | RELAXATION_OPTIMAL | 9        | 127.64    | 0.000       | 9.59%       |
| 10    | 30         | CP_SAT        | OPTIMAL            | 9        | 128.31    | 0.021       | 9.11%       |
| 10    | 30         | HEURISTIC     | FEASIBLE           | 10       | 141.17    | 0.001       | 0.00%       |
| 10    | 30         | LOCAL_SEARCH  | FEASIBLE           | 10       | 141.17    | 0.117       | 0.00%       |
| 10    | 30         | MILP          | RELAXATION_OPTIMAL | 7        | 98.65     | 0.001       | 30.12%      |

| Tasks | Budget (s) | Solver        | Status             | Assigned | Objective | Runtime (s) | Gap to best   |
|-------|------------|---------------|--------------------|----------|-----------|-------------|---------------|
| 10    | 30         | MIN_COST_FLOW | RELAXATION_OPTIMAL | 9        | 127.64    | 0.000       | 9.59%         |
| 25    | 5          | CP_SAT        | OPTIMAL            | 25       | 351.66    | 0.797       | 0.00%         |
| 25    | 5          | HEURISTIC     | FEASIBLE           | 24       | 340.09    | 0.006       | 3.29%         |
| 25    | 5          | LOCAL_SEARCH  | FEASIBLE           | 24       | 341.98    | 0.469       | 2.75%         |
| 25    | 5          | MILP          | RELAXATION_OPTIMAL | 15       | 198.27    | 0.366       | 43.62%        |
| 25    | 5          | MIN_COST_FLOW | RELAXATION_OPTIMAL | 8        | 123.59    | 0.001       | 64.85%        |
| 25    | 30         | CP_SAT        | OPTIMAL            | 25       | 351.66    | 0.956       | 0.00%         |
| 25    | 30         | HEURISTIC     | FEASIBLE           | 24       | 340.09    | 0.006       | 3.29%         |
| 25    | 30         | LOCAL_SEARCH  | FEASIBLE           | 24       | 341.98    | 0.474       | 2.75%         |
| 25    | 30         | MILP          | RELAXATION_OPTIMAL | 15       | 198.27    | 0.375       | 43.62%        |
| 25    | 30         | MIN_COST_FLOW | RELAXATION_OPTIMAL | 8        | 123.59    | 0.001       | 64.85%        |
| 50    | 5          | CP_SAT        | FEASIBLE           | 47       | 643.94    | 5.056       | 0.00%         |
| 50    | 5          | HEURISTIC     | FEASIBLE           | 42       | 580.04    | 0.045       | 9.92%         |
| 50    | 5          | LOCAL_SEARCH  | FEASIBLE           | 42       | 584.09    | 2.729       | 9.29%         |
| 50    | 5          | MILP          | RELAXATION_OPTIMAL | 29       | 383.79    | 0.717       | 40.40%        |
| 50    | 5          | MIN_COST_FLOW | RELAXATION_OPTIMAL | 16       | 241.81    | 0.003       | 62.45%        |
| 50    | 30         | CP_SAT        | FEASIBLE           | 48       | 656.77    | 30.083      | 0.00%         |
| 50    | 30         | HEURISTIC     | FEASIBLE           | 42       | 580.04    | 0.046       | 11.68%        |
| 50    | 30         | LOCAL_SEARCH  | FEASIBLE           | 42       | 584.09    | 2.728       | 11.07%        |
| 50    | 30         | MILP          | RELAXATION_OPTIMAL | 29       | 383.79    | 0.726       | 41.56%        |
| 50    | 30         | MIN_COST_FLOW | RELAXATION_OPTIMAL | 16       | 241.81    | 0.003       | 63.18%        |
| 100   | 5          | CP_SAT        | TIME_LIMIT         | 0        | 0.00      | 2.852       | 100.00%       |
| 100   | 5          | HEURISTIC     | FEASIBLE           | 61       | 831.50    | 0.201       | 1.03%         |
| 100   | 5          | LOCAL_SEARCH  | TIME_LIMIT         | 61       | 840.15    | 5.020       | 0.00%         |
| 100   | 5          | MILP          | FEASIBLE           | 36       | 469.72    | 4.643       | 44.09%        |
| 100   | 5          | MIN_COST_FLOW | RELAXATION_OPTIMAL | 22       | 328.06    | 0.006       | 60.95%        |
| 100   | 30         | CP_SAT        | FEASIBLE           | 60       | 834.53    | 30.166      | 0.90%         |
| 100   | 30         | HEURISTIC     | FEASIBLE           | 61       | 831.50    | 0.197       | 1.26%         |
| 100   | 30         | LOCAL_SEARCH  | FEASIBLE           | 61       | 842.08    | 9.141       | 0.00%         |
| 100   | 30         | MILP          | FEASIBLE           | 43       | 573.15    | 29.504      | 31.94%        |
| 100   | 30         | MIN_COST_FLOW | RELAXATION_OPTIMAL | 22       | 328.06    | 0.006       | 61.04%        |
| 250   | 5          | CP_SAT        | TIME_LIMIT         | 0        | 0.00      | 4.014       | 100.00%       |
| 250   | 5          | HEURISTIC     | FEASIBLE           | 131      | 1,791.04  | 1.654       | 0.92%         |
| 250   | 5          | LOCAL_SEARCH  | TIME_LIMIT         | 132      | 1,807.62  | 5.010       | 0.00%         |
| 250   | 5          | MILP          | FEASIBLE           | 80       | 1,028.17  | 4.538       | 43.12%        |
| 250   | 5          | MIN_COST_FLOW | NOT_APPLICABLE     | 0        | 0.00      | 0.000       | not measured% |
| 250   | 30         | CP_SAT        | FEASIBLE           | 105      | 1,418.65  | 30.308      | 21.62%        |
| 250   | 30         | HEURISTIC     | FEASIBLE           | 131      | 1,791.04  | 1.675       | 1.04%         |

| Tasks | Budget (s) | Solver        | Status         | Assigned | Objective | Runtime (s) | Gap to best   |
|-------|------------|---------------|----------------|----------|-----------|-------------|---------------|
| 250   | 30         | LOCAL_SEARCH  | TIME_LIMIT     | 132      | 1,809.92  | 30.095      | 0.00%         |
| 250   | 30         | MILP          | FEASIBLE       | 77       | 996.01    | 30.019      | 44.97%        |
| 250   | 30         | MIN_COST_FLOW | NOT_APPLICABLE | 0        | 0.00      | 0.000       | not measured% |

| Solver        | Timeouts | Runs | Timeout rate |
|---------------|----------|------|--------------|
| CP_SAT        | 2        | 10   | 20%          |
| HEURISTIC     | 0        | 10   | 0%           |
| LOCAL_SEARCH  | 3        | 10   | 30%          |
| MILP          | 0        | 10   | 0%           |
| MIN_COST_FLOW | 0        | 10   | 0%           |

### Reading the ladder

CP-SAT proves optimality at 10 and 25 tasks and stops doing so as the instance grows, eventually returning nothing at all inside a five-second budget. Those rows are TIME\_LIMIT with zero assignments, which is an honest record: it says the solver found nothing in the time allowed, not that it failed.

The heuristics keep producing usable plans at 250 tasks in about one to two seconds, having placed well over a hundred assignments. That is the central practical result of the whole scalability study and it is not a statement that they are better solvers - it is a statement about what is usable when the shift is already running.

## 18. Constraint Pressure

| Scarcity | Deadline tightness | Feasible | Assigned | Late | Objective | Runtime (s) |
|----------|--------------------|----------|----------|------|-----------|-------------|
| 0.2      | 0.2                | True     | 15       | 2    | 209.91    | 0.0021      |
| 0.2      | 0.5                | True     | 15       | 5    | 209.55    | 0.0022      |
| 0.2      | 0.8                | True     | 15       | 6    | 208.75    | 0.0024      |
| 0.2      | 0.95               | True     | 15       | 6    | 208.33    | 0.0021      |
| 0.4      | 0.2                | True     | 20       | 3    | 282.00    | 0.0055      |
| 0.4      | 0.5                | True     | 18       | 9    | 254.37    | 0.0055      |
| 0.4      | 0.8                | True     | 18       | 11   | 252.85    | 0.0057      |
| 0.4      | 0.95               | True     | 18       | 13   | 253.00    | 0.0054      |
| 0.6      | 0.2                | True     | 23       | 2    | 323.92    | 0.0073      |
| 0.6      | 0.5                | True     | 23       | 8    | 323.98    | 0.0063      |
| 0.6      | 0.8                | True     | 23       | 12   | 322.57    | 0.0062      |
| 0.6      | 0.95               | True     | 23       | 10   | 321.47    | 0.0060      |
| 0.8      | 0.2                | True     | 24       | 3    | 328.83    | 0.0081      |
| 0.8      | 0.5                | True     | 21       | 5    | 286.68    | 0.0100      |
| 0.8      | 0.8                | True     | 24       | 9    | 323.19    | 0.0069      |
| 0.8      | 0.95               | True     | 24       | 13   | 322.15    | 0.0066      |
| 1.0      | 0.2                | True     | 22       | 4    | 299.02    | 0.0104      |
| 1.0      | 0.5                | True     | 22       | 5    | 298.99    | 0.0121      |
| 1.0      | 0.8                | True     | 22       | 9    | 297.71    | 0.0116      |

| Scarcity | Deadline tightness | Feasible | Assigned | Late | Objective | Runtime (s) |
|----------|--------------------|----------|----------|------|-----------|-------------|
| 1.0      | 0.95               | True     | 23       | 15   | 307.33    | 0.0106      |

Resource scarcity and deadline tightness are separate pressures and they degrade the plan differently. Tightening deadlines moves lateness; removing resources moves coverage. Conflating them into a single difficulty dial would hide which one is binding, so they are reported on independent axes.

## 19. Disruption Recovery

| Strategy / severity | Mean service recovery |
|---------------------|-----------------------|
| full_reopt/high     | 0.7%                  |
| full_reopt/low      | 0.6%                  |
| full_reopt/medium   | 0.7%                  |
| local_repair/high   | 0.7%                  |
| local_repair/low    | 0.6%                  |
| local_repair/medium | 0.7%                  |
| none/high           | 0.0%                  |
| none/low            | 0.0%                  |
| none/medium         | 0.0%                  |

  

| Strategy     | Mean replan time (s) |
|--------------|----------------------|
| full_reopt   | 0.0069               |
| local_repair | 0.0059               |

  

| Strategy     | Mean plan churn |
|--------------|-----------------|
| full_reopt   | 14.61%          |
| local_repair | 13.74%          |

Disruption rates of 0%, 5%, 10%, 20% and 30% are crossed with low, medium and high severity, and each case is run three ways: with no replanning at all, with local repair, and with full re-optimization. The no-replanning column is the control - it is what the service level would have been had the system done nothing.

Local repair recovers less service than full re-optimization and is one to two orders of magnitude faster. Which matters more depends on how much of the plan has already been communicated, which is a property of the organisation rather than of the algorithm. No general winner is declared here because the experiment does not support one.

## 20. What-If Analysis

What-if analysis re-runs the real optimization engine over a modified scenario rather than perturbing a stored score. The scenarios are the ones an operator actually asks about: capacity down, demand up, deadlines tighter, a resource gone, travel slower, priorities shifted.

| Scenario                | What changes                       | Question it answers                                |
|-------------------------|------------------------------------|----------------------------------------------------|
| Capacity -20%           | resource capacity scaled down      | how much slack does the plan have?                 |
| Demand +30%             | additional tasks appended          | what happens when volume rises?                    |
| Deadline slack -15%     | deadlines tightened by their slack | which work is already marginal?                    |
| Resource unavailable    | one resource removed for a window  | what is the exposure to a single point of failure? |
| Travel time +25%        | travel matrix scaled               | how sensitive is routing to congestion?            |
| Priority redistribution | task priorities shifted            | does the plan follow the right work?               |

On the demo scenario the deadline-tightening case moved the objective by 1.14. A small delta is a real finding, not a failed experiment: it means the baseline plan was not sitting on its deadline constraints, so tightening them by 15% of their slack changed little.

## 21. Solver Comparison

### *Best objective per instance - and whether it is proven*

| Instance | Budget | Highest objective | Method                         | Objective | Proven optimal? |
|----------|--------|-------------------|--------------------------------|-----------|-----------------|
| ft10     | 30.0s  | CP_SAT            | exact (constraint programming) | 1,400.00  | yes             |
| ft10     | 5.0s   | CP_SAT            | exact (constraint programming) | 1,400.00  | yes             |
| c101     | 30.0s  | HEURISTIC         | heuristic (greedy insertion)   | 1,358.51  | no              |
| c101     | 5.0s   | HEURISTIC         | heuristic (greedy insertion)   | 1,358.51  | no              |
| r101     | 30.0s  | LOCAL_SEARCH      | heuristic (ALNS improvement)   | 1,362.38  | no              |
| r101     | 5.0s   | HEURISTIC         | heuristic (greedy insertion)   | 1,353.86  | no              |

This distinction is the single most important presentation choice in the project. On the job-shop instances MILP reports a makespan far below CP-SAT's, because MILP's big-M formulation relaxes the precedence constraints it cannot represent compactly and therefore reports a lower bound on the makespan. Its plan is re-materialised through the same shared scheduler and passes structural validation - it is a valid schedule - but it is not a schedule any of the solvers proved optimal, and the comparison must not imply otherwise.

### *What each solver is entitled to claim*

| Claim                       | Solvers entitled to it          | Evidence                                                       |
|-----------------------------|---------------------------------|----------------------------------------------------------------|
| Proven optimal              | CP-SAT, when it reports OPTIMAL | 0 mismatches against brute force on 7 fixtures                 |
| Optimal for a reduced model | MILP, MIN_COST_FLOW             | status RELAXATION_OPTIMAL                                      |
| A usable plan               | all five                        | shared independent scorer                                      |
| A complete plan             | none in general                 | scenarios permit unassigned work; coverage is reported per row |

## 22. Failure Analysis

| Category              | Where it appears                                             | Frequency / example                          | Root cause                                         | Mitigation                                                  |
|-----------------------|--------------------------------------------------------------|----------------------------------------------|----------------------------------------------------|-------------------------------------------------------------|
| Solver timeout        | CP-SAT and LOCAL_SEARCH at the top of the scalability ladder | CP-SAT 20%, LOCAL_SEARCH 30% of runs         | no plan or an unproven one at the budget allowed   | raise the budget, or use a heuristic where seconds matter   |
| Relaxation limitation | MIN_COST_FLOW on every routing instance                      | 46 of 99 customers served on c101            | no route ordering and no time windows in the model | use it to bound a relaxation gap, never as a routing answer |
| Heuristic gap         | constructive heuristic under heavy scarcity                  | worst shortfall against the reference -3.70% | greedy density order is myopic about dependencies  | local search from that start, or a longer budget            |

| Category              | Where it appears                                | Frequency / example                                     | Root cause                                                   | Mitigation                                                  |
|-----------------------|-------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|
| Big-M relaxation      | MILP where precedence is binding                | reports RELAXATION_OPTIMAL, not OPTIMAL                 | precedence expressed with a large-M disjunction              | treat the figure as a bound, verify the plan structurally   |
| Constraint bottleneck | deadline tightness in the pressure suite        | lateness rises as slack is removed                      | release-to-deadline windows leave no slack for travel        | widen windows or add resources before tightening further    |
| Resource bottleneck   | capability scarcity in the pressure suite       | coverage falls as scarce capabilities are removed       | a capability with no eligible resource cannot be substituted | duplicate the scarce capability, or retrain                 |
| Infeasible input      | any solver, on a scenario with no eligible pair | status INFEASIBLE or NOT_APPLICABLE with zero objective | demand exceeds capability, capacity or horizon               | validate before solving; the binding constraint is reported |

Every category above was observed in the recorded runs rather than anticipated. Two of them - the relaxation limitation and the big-M relaxation - are permanent properties of the methods, not bugs, and are reported on every run rather than treated as surprises.

Integrity of the live set: no inconsistencies found.

## 23. Product Architecture

The product surface is deliberately thin over the engine. Each layer reads the core through the same public entry points, so the CLI, the HTTP API and a benchmark run cannot diverge in behaviour.

| Layer       | Responsibility                                             | Reads from core                                     |
|-------------|------------------------------------------------------------|-----------------------------------------------------|
| CLI         | scripted, reproducible operation and evidence inspection   | Planner, Replanner, SimulationEngine                |
| API         | interactive operation over HTTP with typed schemas         | same entry points, Pydantic schemas at the boundary |
| Persistence | durable scenarios, plans, disruptions, events, experiments | domain objects, stored relationally                 |
| Frontend    | the operator's view of the same state                      | the API only, never the database directly           |

API schemas are kept separate from domain entities. The domain is a scheduling model that predates any transport, and binding it to a wire format would make the engine impossible to use from a test without a web server.

## 24. User Workflow

The intended path through the product, and the path the demo walks:

| Step                   | What happens                                                 | Evidence produced             |
|------------------------|--------------------------------------------------------------|-------------------------------|
| 1. Open scenario       | inspect resources, tasks, capabilities, windows, maintenance | validation report             |
| 2. Validate            | static feasibility and the binding constraint                | feasibility status            |
| 3. Optimize            | solve, re-score centrally, validate the plan                 | plan with solver runs         |
| 4. Inspect assignments | task, resource, window, route, reason                        | per-assignment explanations   |
| 5. Simulate            | discrete-event execution of the plan                         | ordered event log             |
| 6. Disrupt             | apply a disruption, identify consequences                    | affected task set             |
| 7. Local repair        | re-insert affected work, keep the rest                       | repaired plan, churn, runtime |

| Step                    | What happens                                         | Evidence produced        |
|-------------------------|------------------------------------------------------|--------------------------|
| 8. Full re-optimization | re-solve the disrupted scenario                      | plan, objective, runtime |
| 9. Compare              | objective, service, churn, runtime, recovery         | comparison table         |
| 10. What-if             | re-solve a modified scenario                         | delta versus baseline    |
| 11. Export              | plans, metrics, timelines, manifests with provenance | artifacts                |

On the recorded run the disruption identified 7 affected tasks, and the demo reports the recovery, the churn and the runtime for both strategies before drawing any conclusion about which is preferable.

## 25. Reproducibility

| Suite                   | Experiment                                                | Commit   | Recorded (UTC)            |
|-------------------------|-----------------------------------------------------------|----------|---------------------------|
| constraint_pressur<br>e | benchmark-constraint_pressure-20260926T232436Z-bf748adf7e | 5fe96854 | 2026-09-26T23:24:36+00:00 |
| disruption_stress       | benchmark-disruption_stress-20260926T232436Z-bf748adf7e   | 5fe96854 | 2026-09-26T23:24:36+00:00 |
| scalability             | benchmark-scalability-20260926T230011Z-bf748adf7e         | dc70e26f | 2026-09-26T23:00:11+00:00 |
| solver_comparison       | benchmark-solver_comparison-20260926T230445Z-bf748adf7e   | dc70e26f | 2026-09-26T23:04:45+00:00 |
| time_budget             | benchmark-time_budget-20260926T232445Z-bf748adf7e         | 5fe96854 | 2026-09-26T23:24:45+00:00 |

### How to reproduce

```
python -m orion verify # brute-force reference agreement
python -m orion demo # deterministic end-to-end run
python -m orion benchmark --config configs/benchmark.yaml
```

### What is and is not deterministic

Scenario generation, planning, simulation and disruption application are deterministic given a seed: the same seed reproduces the same scenario, the same assignments, the same objective and the same validation result. Runtime does not reproduce, and neither does anything derived from it - which is why runtime columns are measured rather than asserted.

Experiment identifiers and timestamps necessarily differ between runs. That is the only intended difference; the rows beneath them should match.

Two solver results are deliberately not claimed to be deterministic: an exact solver's answer is, but wall-clock budgets mean a run that reaches TIME\_LIMIT can return a different plan on a different machine. The status is reproducible; the plan at a tight budget is not.

## 26. Limitations

### Method limitations, stated as properties of the methods

MIN\_COST\_FLOW is a transportation relaxation. It has no route ordering and no time-window feasibility, so it serves fewer customers than any routing solver and its objective is not comparable to theirs as a quality measure. It is useful as a bound and for very large instances, not as a routing answer.

MILP relaxes precedence. Precedence is expressed with a large-M disjunction, so where precedence binds, MILP's objective is a bound rather than an achievable value. It reports RELAXATION\_OPTIMAL accordingly.

CP-SAT does not scale here. It proves optimality on small instances and returns nothing at all inside a short budget on large ones. That is a finding, not a failure to hide.

The heuristics are incomplete by construction. They may leave work unplaced where an exact solver would not, and their gap grows with constraint pressure.

### ***Evidence limitations***

Two public instances families are benchmarked end to end. The available 100-customer Solomon set and four job-shop instances are a sample, not a survey.

No job-shop optimum is claimed. The downloaded artifact does not carry verified optimal makespans, so none are quoted.

Runtime figures come from one machine and one OR-Tools build. They are comparable within this report and not a claim about other hardware.

The disruption study uses synthetic disruption generators. Real failure distributions are not represented.

Scenarios permit unassigned work, so coverage must be read per row. A high objective does not by itself mean full coverage.

### ***Scope***

ORION is a research and demonstration system. It has not been deployed in production, has not been validated against operational data from a live organisation, and makes no claim of universal solver superiority. Its scenarios - logistics, maintenance, emergency response, infrastructure and fleet scheduling - are illustrative of the problem class, not descriptions of any specific operation.

## **27. Conclusion**

The engineering question has a partial but concrete answer. Exact optimization is achievable and verifiable on small instances - 7 brute-force fixtures with 0 proven mismatches - and stops being achievable as instances grow. Heuristics remain usable throughout the range tested, at a measured and non-trivial quality cost. Recovery from a disruption is fast and largely effective when it is local, and slower but marginally better when it is global.

On the recorded demo run, local repair recovered 92.5% of the lost service in 0.090 s against full re-optimization's 5.024 s. The most useful output of the project is not a number but the shape of that trade-off, measured rather than asserted, with its own failure modes documented.

The remaining work is product depth and broader evidence, not new optimization machinery: more benchmark instances, richer scenario realism, and a deployed frontend over the API. The optimization core is frozen against further change except where a reproducible correctness failure requires it.